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Palaeomagnetism of the British Tertiary Igneous Province: The Skye Lavas

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Summary

Sampling of five lava sequences from different Skye lava sets as put forth in the *Memoir of the Geological Survey of Scotland*, The Geology of Northern Skye, has produced the following information.

(1) Each of the independently proposed Skye lava sets has a different and characteristic distribution of directions of magnetization. Separate lava sequences within any one set have so far shown similar distributions. It follows that the sets were emplaced at different times, although we could not discover the order from the palaeomagnetic evidence alone.

(2) There is some conflict between palaeomagnetic and geological inference concerning the time of tilting of the lavas. The geological evidence seems stronger—that the tilts are original, and should not be corrected for.

(3) As in earlier British Tertiary studies, all lavas are reversely magnetized, and many of the lavas have that shallow reversed inclination which characterizes the British Tertiary igneous province.

1. Introduction

This investigation furthers researches into the palaeomagnetism of the British Tertiary Igneous Province. We have sampled chronological sequences of lavas from four of the five possible Skye lava sets proposed by Anderson and Dunham in the Geology of Northern Skye, a *Memoir of the Geological Survey of Scotland* (1966). One of the objectives was to examine the palaeomagnetic relationships from set to set. Another was to test whether the lava tilts were depositional or post-depositional. We have still not found a single British Tertiary lava which is normally magnetized, and so we are also pursuing the phenomenon of field behaviour over an apparently quite short time interval. Our sequences continue to provide 'groups' of consecutive lavas with closely similar directions of magnetization, as in Northern Ireland (Wilson 1970) and in Hawaii (Doell 1969).

2. Collection and measurement information

Appendix I contains detailed information of the locations of, and geological relationships between, the collecting sites. The five lava sequences are labelled J, K, L, M and N (within our larger British Tertiary collection). Fig. 1 shows our

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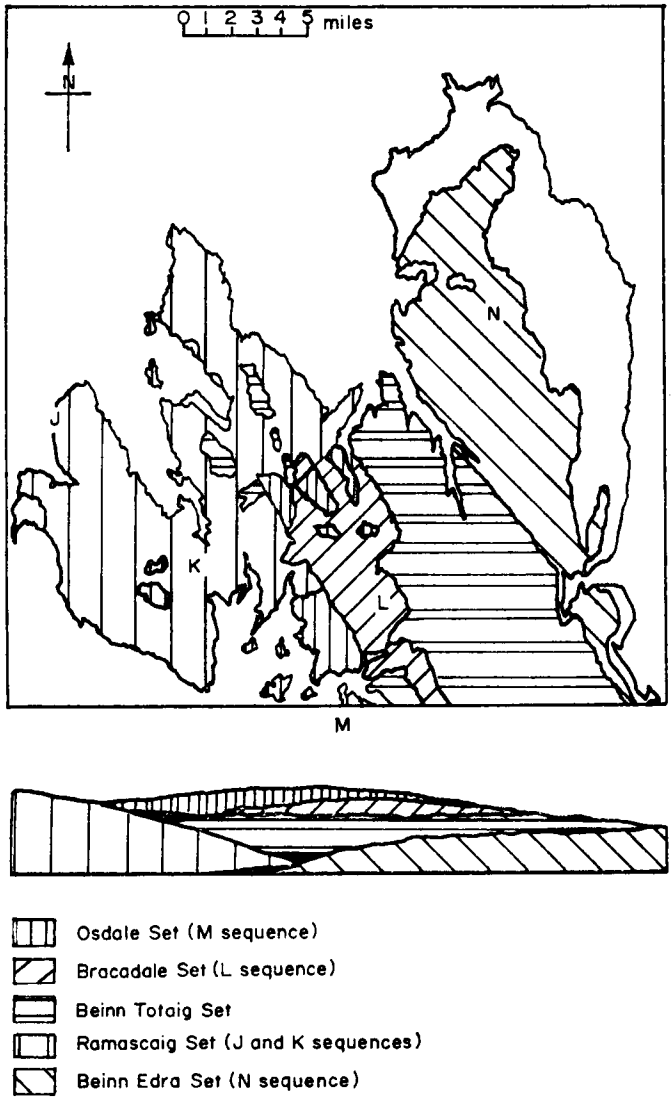


FIG. 1. Map of Skye collection sites and of the Memoir's lava sets. Based on Crown Copyright Geological Survey map. Reproduced by permission of the Controller of H.M. Stationery Office.

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sites of collection and the five lava sets proposed as being of different ages in the Memoir of Northern Skye. Our sequences are related to these lava sets as follows (the numbers are the time order suggested in the Memoir, 1 being oldest):

Sequence	Lava set
J and K	2, Ramascaig set.
L	5, Osdale set.
M	4, Bracadale set.
N	1, Beinn Edra set.
Note sampled	3, Beinn Totaig set.

We found, as did Khan (1959), that a large fraction of the hillside lavas had been struck by lightning. Many such specimens had very strong and random magnetizations. Much of the already short L-section especially had to be rejected for this reason. In addition, a few low-lying lavas were rejected because we later became uncertain whether or not they were still *in situ* (K and M sections particularly). There then remained 90 lavas from all five sequences. The palaeomagnetic details are reported in Appendix 2.

3. The tilts of the lavas and the palaeomagnetic relationships between the Memoir's five lava sets

Fig. 2 displays the individual lava-mean directions. The upper row of diagrams show this information with tilt-correction; that is, after restoring the lavas to a horizontal attitude, as is usual. Anderson is strongly certain (in the Northern Skye Memoir) that the lava tilts are depositional, because of lack of tilting in the underlying sediments. We have therefore also displayed, in the lower row of diagrams in Fig. 1, our lava-mean directions without tilt correction.

The first fact of note is that lava sequences J and K have very similar groupings of lava directions while sections M and N are quite distinct from each other and from (J+K)*. This is completely in accord with the lava sets proposed in the Memoir. Sequences J and K come from the same lava set and seem on the palaeomagnetic evidence to be closely contemporaneous. It is also worth noting that sequence N gives a result which agrees closely with that which Khan (in his thesis (1958)) found from lavas from the Quirang and from the Storr sequence which lie also in lava set 1. We therefore have here a second indication that each set erupted at different times but that during the deposition of any one set there was a characteristic distribution of magnetic field vectors.

The second topic of discussion is the problem of lava tilts raised earlier. Were the lavas originally flat-lying, or tilted as we find them today? The traditional palaeomagnetic approach is to look at the dispersion of a set of measurements before and after tilt-correction. The set with the least dispersion is taken as representing the original attitude of the lavas (or sediments—this is Graham's original fold test (1949)). This can be applied in three different ways in the present case.

(*a*) From Fig. 2(*a*) it is obvious that lava sequences J and K coincide better after tilt correction. The before-tilt standard deviation of 11·25°, reduces to 10·46° after tilt correction. This piece of evidence supports post-eruption tilting, but is not strong enough to be conclusive.

(*b*) Corresponding to Fig. 2(*d*) the overall standard deviation of the 90 lavas is 13·60° before tilt correction, and 14·31° after tilt correction.

* Sequence L contained too few lavas to be of use in this context.

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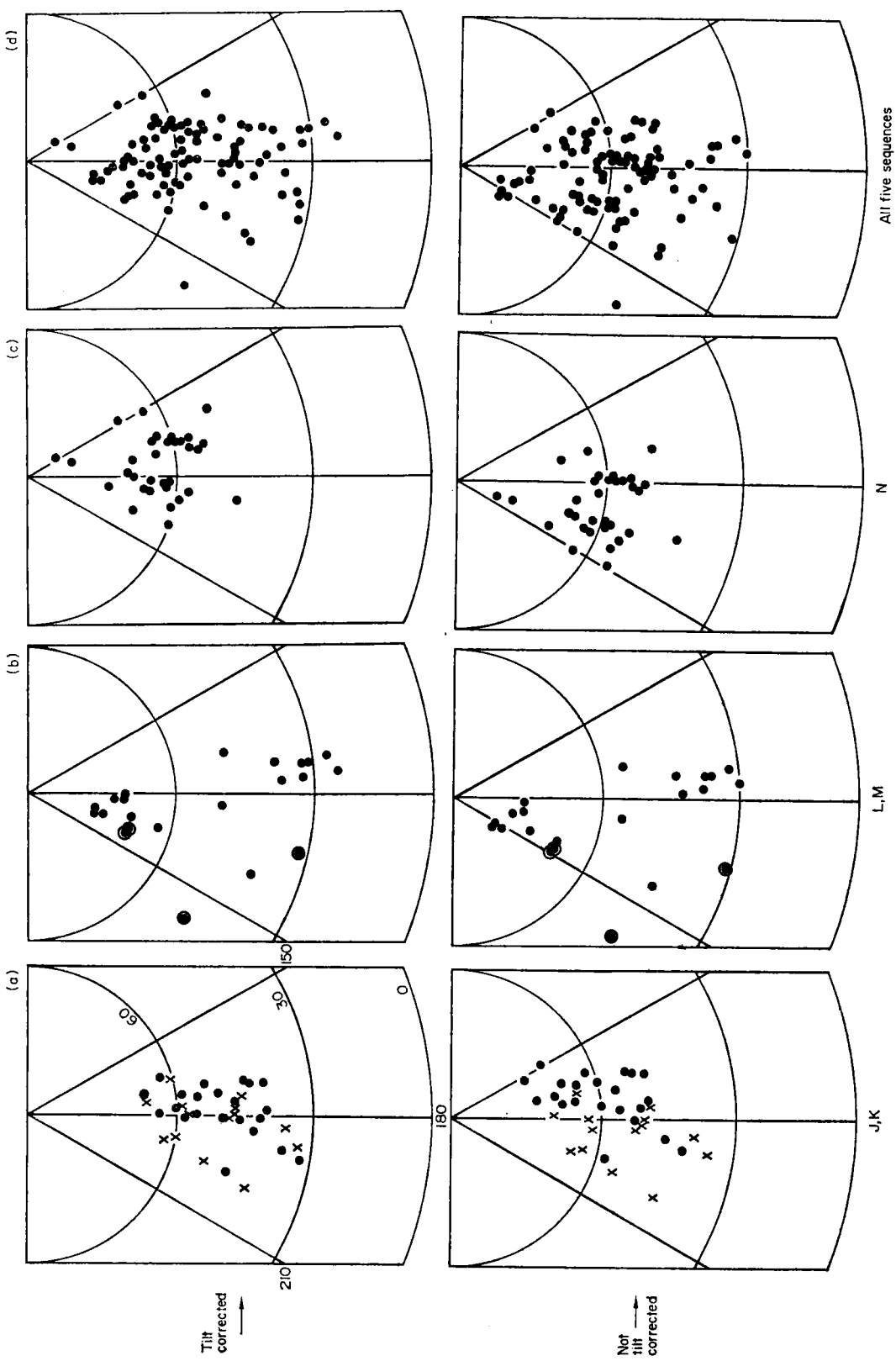


FIG. 2. Lava-mean directions from Skye. Upper row of figures contains tilt-corrected data. Lower row of figures contains non-tilt-corrected data.

FIG. 2(a) Lava sequences J (●) and K (×). FIG. 2(b). Lava sequences L (●) and M (●).
FIG. 2(c). Lava sequence N (●). FIG. 2(d). All five lava sequences.

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Taken at face value, this evidence supports the opposite view that the present lava tilts are original. However, it is weakened by the peculiar overall distribution, which is far from Fisherian. Whereas there is some reason for supposing (in (*a*) above) that J and K sequences *ought* to coincide since they may be contemporary there is no compelling reason for supposing that (J+K), M, and N sequences *ought* similarly to do so. They obviously do not coincide with each other either before *or* after tilt correction.

(*c*) Some groups of three or more consecutive lavas have closely grouped directions of magnetization. These groups, regardless of the lava sequence they come from, are contained in the encircled areas of Fig. 3. It is clear that despite the *increase* of the overall standard deviation after tilt correction in (*b*) above, these lava *groups* become more tightly organized when tilt corrected. The word ' organized ' is chosen deliberately, because there is not a simple aggregation of vectors, but rather a pattern appears.

(*d*) The entire lava area is dissected by sets of faults in different directions. Some, post-eruptive tilting would be expected to be associated with this faulting but its magnitude is unclear.

It is not easy to draw a simple conclusion. First, we are not sure of the meaning of lava groups like these. Either they mean that the groups of lavas erupted swiftly with respect to secular field variation like that of today, or else they mean that the field itself was for certain periods relatively quiescent—during which time a normal rate of

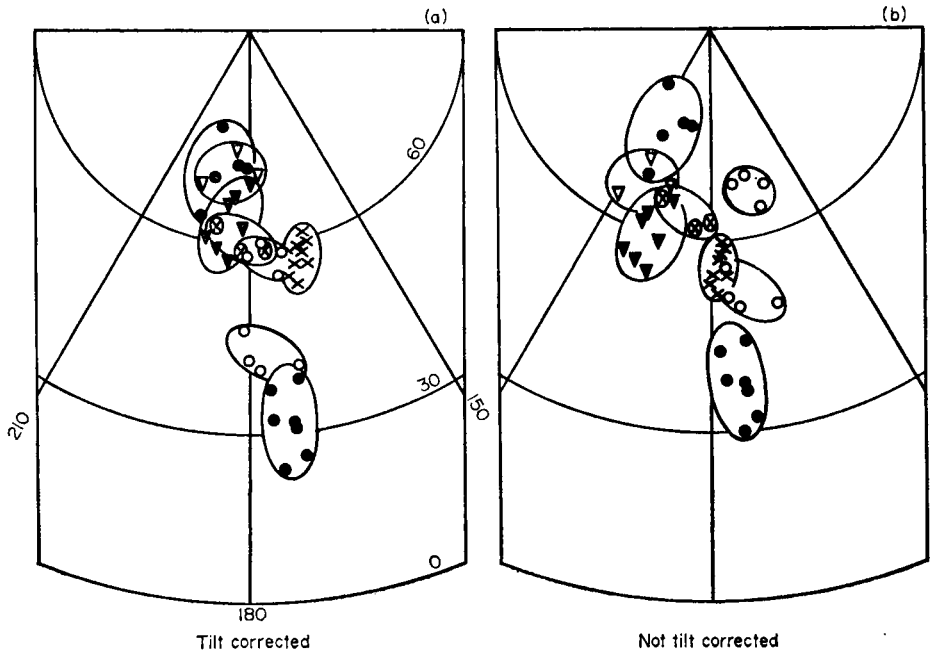


FIG. 3(a). Tilt-corrected lava groups, each group being encircled and marked with a separate symbol.

FIG. 3(b). Non-tilt-corrected lava groups, as in Fig 3(a). Symbols are as follows ○ J8–11 (steeper) and J12–15 (shallow inclinations); ⊗ K10–12; ● M12–16 (steeper) and M17–23 (shallow); ▼ N17–23; × N26–35 (omitting 30 and 31 which straddled a slight fault and may have been affected by it); ▽ N36–38.

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eruption was being pursued. Second, why should these lava groups (in Fig. 3) have the observed geometrical relationship, which is certainly not simple, yet has some form to it?

It is possible that the tightness of the tilt-corrected lava groups and the J–K relationship mean that the tilt is post-eruptive; but this opposes the geologically-based conclusion in the Memoir. The palaeomagnetic evidence is not strong enough to be utterly conclusive, and Anderson's evidence for depositional tilt probably still holds good.

The only positive conclusions of this section are that the various lava sets were erupted at different times as Anderson and Dunham suggested; and that from place to place laterally within any one set, the distributions of directions of magnetization have so far matched stratigraphically.

4. Shallow reversed magnetizations

In two other publications, one about the Mull dykes (Ade–Hall *et al.* 1972) and one about N. Irish lavas (Wilson 1970) we have found many shallow reversed directions of magnetization. Here again we find them in Skye. A comparison of the present Fig. 2(*d*) with Fig. 2 of Wilson (1970) and Fig. 8 of Ade–Hall *et al.* (1972) shows the similarities. Clearly they are a common feature in the British Tertiary igneous province, and we must attempt to interpret them. However, there is yet considerably more data from Skye, Arran and Lundy dykes and from Mull lavas to be published, so that it would seem best to await a complete review of all the evidence. For now we may say that Storetvedt (private communication) thinks that there is considerable secondary magnetization influencing these directions, and that the Eocene field inclination was about $-40°$. The Skye data does not support this interpretation, both because $-40°$ is inconsistent with this data, and because the shallowest lava directions are not a function of depth of burial (many of the highest lavas have shallow directions). Further, the stratigraphic correlation carried out in Antrim (Wilson 1970) would not have been possible if secondary magnetization (a highly variable quantity) had interfered with primary directions. However, the last word is yet to be said.

5. The overall distribution of directions of NRM

A glance of Fig. 2 convinces us that no single lava sequence describes the representative palaeomagnetic field distribution of Eocene times in Britain. Each sequence has its own characteristic and distinguishable set of NRM directions. It seems equally sure that all the sequences together also do not define an average field. The distributions in Figs 2(*d*) and 3 are not simple enough to permit us to calculate a meaningful overall mean direction. Of course the results mean something interesting, but they cannot be used for purposes of continental drift.

Presumably we are catching a great deal of detail about field behaviour, at the expense of not having an 'average' look. The similarity of the Mull (Ade–Hall *et al.* 1972), N. Ireland (Wilson 1970) and Skye results suggests moreover that what we are seeing (especially the shallow reversed directions) is shared throughout the British Tertiary igneous province. In a few years, we shall have considerably more information than now, and it will then be sensible to try to formulate a strong interpretation of all the results together, both in terms of magnetic field behaviour and perhaps in terms of stratigraphic correlation throughout the province.

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Appendix 1

Locations of lava sequences sampled in Skye

Sequence code	Bottom of sequence	Top of sequence	Order of deposition
J (map NG15NE)	J55 at NG1606–5555	J70 at top of cliff NE1613–5542 and south along cliff edge to J78 at NG1586–5500	2
K (map NG24SW)	K3 at NG2374–4497	K22 at NG2226–4460 (on Mcleod's Table North	2
L (map NG34SE)	L1 at NG3671–4078	L8 at NG3658–4167 (on Benn-na-Cloiche)	5 (youngest)
M (map NG33SW)	M7 at NG3369–3050	M27 at NG3394–3000	4
N (1″ sheet 24)	N1 at NG466–592 (to nearest 100 m)	N39 at NG465–589 (to nearest 100 m)	1 (oldest)

Four figure references are to nearest 10 m.
Three figure references are to nearest 100 m.

Appendix 2

		Tilt corrected		Not tilt corrected		
LAVA	N	DECL	INCL	DECL	INCL	ALPHA 95
J055	2	163·9	−62·1	148·2	−69·3	3·1
J056	4	171·3	−43·8	165·3	−52·2	3·0
J057	4	187·6	−36·5	185·4	−46·2	2·7
J058	3	189·1	−32·2	187·4	−42·0	1·4
J059	4	175·8	−47·5	169·8	−56·3	1·0
J060	5	170·6	−45·0	164·1	−53·4	1·8
J061	4	183·9	−43·2	180·3	−52·7	2·2
J062	4	173·7	−55·1	169·8	−56·3	3·3
J063	4	172·3	−58·4	164·1	−53·4	1·4
J064	4	177·2	−59·9	180·3	−52·7	2·3
J065	4	180·8	−58·0	164·7	−63·7	0·9
J066	4	178·5	−40·6	174·3	−49·7	2·1
J067	4	181·3	−46·6	176·6	−55·8	2·3
J068	4	180·4	−41·9	176·3	−51·1	3·2
J069	4	171·9	−40·9	166·6	−49·4	2·4
J070	3	169·7	−65·8	152·2	−73·5	1·3
J071	4	179·3	−55·6	171·8	−64·7	2·7
J072	4	179·1	−63·4	167·7	−72·4	1·3
J073	4	180·6	−50·1	175·0	−59·3	4·3
J075	4	169·8	−53·4	160·6	−61·5	2·6
J076	4	173·1	−51·0	165·5	−59·6	2·0
J077	4	195·8	−47·9	194·5	−57·8	3·1
K003	4	174·0	−66·0	178·7	−69·6	3·4
K004	4	188·6	−59·8	193·6	−62·7	5·9
K005	4	165·9	−60·3	168·4	−64·1	2·8
K007	4	174·6	−46·0	176·6	−49·5	2·1
K008	4	180·9	−49·1	183·7	−52·4	4·1
K009	3	182·5	−36·5	184·3	−39·8	3·3
K010	4	180·7	−58·0	184·5	−61·3	1·3
K011	4	190·1	−62·1	195·8	−64·9	1·8
K012	4	176·7	−58·8	180·3	−62·3	1·3
K013	4	179·5	−47·8	182·1	−51·2	2·2
K014	4	186·5	−33·3	188·2	−36·3	3·0
K015	4	198·6	−42·9	201·6	−45·4	3·0
K021	3	177·9	−47·4	180·3	−50·8	1·6
K022	3	194·5	−53·0	198·7	−55·6	5·8
L001	4	201·3	−68·8	208·1	−67·5	2·4
L002	4	192·4	−31·9	194·2	−31·2	2·6
L003	4	200·1	−68·6	206·9	−67·3	2·3
L006	3	218·7	−48·9	221·2	−47·0	5·3
M007	4	183·7	−72·6	194·9	−77·8	3·0
M008	4	195·6	−74·3	213·1	−78·7	1·6
M010	4	168·0	−49·2	169·2	−55·2	1·1
M011	4	183·5	−50·3	186·9	−55·8	5·0
M012	4	195·8	−76·0	216·5	−80·2	3·7
M013	4	194·8	−62·8	202·9	−67·5	2·6
M014	4	183·6	−70·7	193·2	−76·0	4·3
M015	4	182·3	−70·5	191·3	−75·3	4·1
M016	4	193·0	−68·6	203·9	−73·3	4·0
M017	4	173·5	−31·2	174·4	−37·0	2·6
M018	4	176·9	−37·4	178·4	−43·1	2·2
M019	4	176·4	−32·6	177·6	−38·4	3·8
M020	4	173·2	−38·5	174·5	−44·3	4·4
M021	4	175·6	−24·4	176·4	−30·2	3·6
M022	3	172·6	−26·7	173·4	−32·5	5·2
M023	4	173·6	−32·0	174·6	−37·8	5·9
M024	4	200·0	−41·0	203·8	−45·5	3·9
M026	3	193·7	−76·0	213·6	−80·4	5·4

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N001	4	159·3	−50·5	167·7	−50·0	2·7
N002	4	150·7	−63·0	164·5	−63·3	3·8
N003	4	148·2	−68·5	166·2	−68·8	3·5
N004	4	164·2	−63·8	177·8	−62·4	4·1
N005	4	183·6	−61·8	194·3	−58·4	3·1
N006	4	199·1	−59·8	207·1	−55·0	4·3
N007	3	183·0	−61·4	193·6	−58·0	4·5
N011	4	181·7	−64·8	194·0	−61·5	2·0
N012	3	170·0	−63·4	182·9	−61·4	4·1
N016	4	162·5	−62·5	175·5	−61·4	3·1
N017	3	185·6	−57·0	194·4	−53·5	4·1
N018	4	183·0	−62·0	193·8	−58·6	1·4
N019	4	192·3	−60·2	201·3	−56·0	0·6
N020	4	186·9	−65·3	198·7	−61·4	2·7
N021	3	180·3	−68·5	194·9	−65·3	1·8
N022	4	189·0	−58·8	198·0	−55·0	1·8
N023	4	186·3	−66·0	198·5	−62·2	1·8
N025	3	146·5	−83·3	199·6	−81·5	2·7
N026	3	171·1	−54·9	180·3	−53·0	2·2
N027	4	166·7	−58·4	177·5	−56·9	1·4
N028	4	166·1	−59·0	177·2	−57·6	2·7
N029	4	169·5	−53·5	178·4	−51·8	2·2
N030	3	171·3	−68·3	187·0	−65·9	3·1
N031	4	186·7	−46·9	192·8	−43·4	2·1
N032	4	169·6	−56·5	179·5	−54·7	0·7
N033	4	166·4	−56·3	176·5	−54·9	2·0
N034	4	165·0	−60·6	176·9	−59·2	1·4
N035	4	164·7	−59·7	176·3	−58·4	1·4
N036	4	197·9	−67·7	208·9	−62·9	0·9
N037	4	178·2	−69·7	193·9	−66·6	3·8
N038	4	187·0	−73·6	204·2	−69·5	0·7
N039	4	161·5	−80·7	197·5	−78·2	2·0

N = number of separately oriented specimens used.
DECL = declination of lava-mean magnetization.
INCL = inclination of lava-mean magnetization.
ALPHA 95 = radius of 95 per cent of confidence.

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